Question 23 (10 marks) Use the Question 23 writing booklet.

A messaging system allows users to exchange text messages online. User details and messages are stored in a flat-file database. A record from the flat-file database is shown.

SenderPhone	0312 621 981
ReceiverPhone	0815 683 665
Message	Meeting is at 10 am
MessageTime	31/01/2023 16:17
MessageRead	☑

Both the senders and receivers are users of the messaging system.

(a) Copy the following table into your writing booklet. **3**

In your table, recommend an appropriate field size and data type for SenderPhone and MessageRead. Justify each choice of data type.

Field	*Field size*	*Data type*	*Justification for choosing the data type*
SenderPhone			
MessageRead			

(b) The flat-file database is to be converted into a relational database. In addition to the phone number, more information such as name and email address will be stored for each user. **3**

Draw a schematic diagram to show how the database can be normalised. Clearly label the names of the tables, primary and foreign keys and any relationships.

(c) The messaging system will be further enhanced to allow users to send images and attach files. **4**

Describe ethical and technical issues that may result from this further enhancement.